

SecureCode AI

M-A2026 Academic Snapshot

Reproducible academic experiment report

Problem statement

Can an evidence-gated pipeline detect CWE-89, propose a bounded repair, validate it in an ephemeral workspace, and report the result without changing the source checkout?

Solution

Language adapters feed a neutral ProgramGraph, normalized signals, EvidenceGraph, verdict, repair proposal, ephemeral validation, and source-free reports. The Docker oracle isolates corpus execution; the local model is restricted to literal loopback.

Method

Evidence combines a pinned public Python demonstration, a 312-cell five-configuration development matrix, one real local-model instructor run, and the canonical offline repository quality gate.

Quality record

Canonical quality: PASS; 2912 passed; 32 skipped; 80.14% Core branch coverage.

Base implementation commit: e33feb44e052a47e12e13424974dadd1014da2ed

Bound subject: 1f7bcc8621558ab9ecf8d9839163b373b877dbe53f665b75159fd6eed328e541

Experimental results

Development matrix: 312 of 312 cells recorded; 171 failed cells; aggregate incomplete.

Deterministic-only: TP=1, FP=0, TN=12, FN=11, precision=1.00, recall=0.0833, F1=0.1538.

One-shot local model: TP=9, FP=3, TN=24, FN=27, precision=0.75, recall=0.25, F1=0.375;

24 structured-output failures.

Repair study: 0 attempted patches, 0 independently validated patches, root-cause repair rate unavailable.

Real local-model demonstration

Outcome=COMPLETED; lanes=AGREED; patch=PROPOSED; ephemeral validation=PASSED; source unchanged=true.

Observed Ollama 0.16.2; model qwen2.5-coder:7b-instruct-q4_K_M; quantization Q4_K_M;

digest sha256:dae161e27b0e90dd1856c8bb3209201fd6736d8eb66298e75ed87571486f4364.

Conclusion

The pinned Python case demonstrates the intended detect, suggest, validate, and report flow. The matrix does not support general accuracy or repair-effectiveness claims because recall is low, 171 model cells failed, and no patch entered the repair study.

This is an academic prototype snapshot. It does not claim G7, G9, v1.0, PROJECT CLOSED, production readiness, or release readiness.

Delivery status

NOT_READY

- Instructor reference, individual-work approval, exact defense slot, and upload deadline remain unconfirmed.
- This academic snapshot does not claim G7, G9, v1.0, PROJECT CLOSED, or release readiness.
- The development benchmark retains incomplete model cells and no repair-rate claim.
- Independent reviews and protected delivery remain pending.

Data handling: aggregate facts, paths, and hashes only. Raw source, prompts, model responses, credentials, and secrets are omitted.

Evidence inventory

README.md 1c6107e484b35f7edcea175a8f0aab51cab9c1515306d0f02f00ca1e6cbb11a

artifacts/gates/G2/promotion-manifest.json d63e16e891971b7f75d02a475716b323b0f92437fab0b006c5ffcd6b724f07e2

artifacts/gates/G2/test-results/deterministic-core-validation.md

dd64d5feff2b62ddf9b3244cddb0e87da22d7b5e1c569a27032d71e80eb2264ce

artifacts/gates/G3/promotion-manifest.json 012d85d83108e308cbf16801005a65b022859c5199a3d6f21355edf3b27ffd2d

artifacts/gates/G3/test-results/g3-validation.md 0b1d8ffdd2f4ef5d57bec2d9913162c9e3422d9d0d277baebb6c7dd150625eb9

artifacts/gates/G4/promotion-manifest.json c59d5428d8f068417da21064bac0cd98ac81f43c266ca116f3292765476ad5bb

artifacts/gates/G4/test-results/g4-validation.md 41bf0550780127718aec1f6cbe2862347cb3515d8c63411361cc40fa25686da4

demo/mvp_cwe89_demo.py 90babcd217c88199d559a54bfc4e10871fa2eb224c1979dfc1720e61f8e1a1a

demo/p917_real_local_demo.py 2550ce04ca5e39a3616fef5e3db730537c52794869345b9405bd355b264c4275

docs/TEACHER_QUESTIONS.md 9265f214a84fc0187baf98a702a2a3f65c03927c3d55785ce26f44f037b9dddf

evaluation/development/corpus-manifest.yaml 59ff6effc451cea622c258884daeb71582f52137e986b4eaf0e5f28d421202bf

evaluation/development/results/aggregate.json deae563fe08580a2ed3dc447ba19d020f31e63797eb8b83c8b6737ad3a674407

evaluation/development/results/recomputed.json deae563fe08580a2ed3dc447ba19d020f31e63797eb8b83c8b6737ad3a674407

evaluation/development/results/run-records.jsonl 396667fc4e2a08046a4d487ed6306f670c76a356800e2ead5b56ebf047790be

evaluation/development/run-plan.yaml fa294da989a423167f5a6bb04e693b4926b414d4c557bc1250b27b420d3b7728

packages/adapters/src/securecode_ai/adapters/cwe89_multilanguage.py

57aca1d1ab5a5b91bb4792026467ad6f12306c70da49f0ac748cf779c3f6b0b9

packages/adapters/src/securecode_ai/adapters/cwe_portfolio.py

49cfbda8da069575a20212c37ab66fad48542ee040d206cbde42e06d137754ad

packages/adapters/src/securecode_ai/adapters/program_graph.py

a0b325ea18385156a6bab2ba1ef7977c94656ae05defaa57a3a1a5ccf4507e80

pyproject.toml 0988a050872ff202c3575146313273426736112b5e80080832f2537d5cf52cc8

report/development-benchmark/README.md df7b813d7e330c04bb05bfb62dbe9ead8addabb5de84649543505ed2dac5d15

report/development-benchmark/limitations.md 2ec663f5f21ec7461b29a93bab8bea509c5aedb463797699b5ebc8d82f73756e

report/m-a2026/evidence/p917-real-local/integrator-receipt.json

7fa7d9f106abdc8c991df34c25b43bf5fcd94687eebc5fe313667d061c540603

report/m-a2026/evidence/p917-real-local/p917-ephemeral-validation.json

130c1595e7e5aee1847a5c7e79422efef2e685456f00f9c67034b310e2a89bd8

report/m-a2026/evidence/p917-real-local/p917-local-demo.html

400ff949e73e65f6b6987551e75f8c105b47b95f83b69d648bcbcabed220c6be

report/m-a2026/evidence/p917-real-local/p917-local-demo.json

d18a2b706dd4faae55c4139a464a50c1de51401718cd95a89bed7047f651ec6b

scripts/build_m_a2026_submission.py 0bc2dd2cec09dd2f47aa82be72135174cf1e5cd689c30e3cdfdafd233bad4edc

tests/integration/test_cwe_portfolio_pipeline.py e5b3fae0abb313477c7b74379471a7bc887c3043c8e9fac48bbea1156aa3d6a7

tests/integration/test_m_a2026_submission.py 5f46ea984efd26a855d73a19d2b7b1eab585de8b4abd2fd588e4b2fca5494598

tests/integration/test_multilanguage_pipeline.py 00eb71e74e88dfe4b5a424f44acc9f0da0c81dddac6c59a743f013d17dc33fc

tests/integration/test_mvp_demo.py 2faa6dc45ebd7cb5719f27c90a778f8068cc8cf27f62ec97e7d61e10f3d84dfa

tests/integration/test_p917_real_local_demo.py 91729ef1916df05e502d25bbb6e3b7b54955f804bd0eebc6a044fd8a7312b21a

tests/unit/test_cwe_portfolio.py 738e0fd7b7752c59faac5d937272233c8248f90a9e2ab7539b3370bbbe85faf6

uv.lock 019129e771ddddd3aa83d1feeadd60a6f2b19945baa187e0a171ff32b5aa8139